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Well intervention disconnection system

Abstract

An intervention system is provided for a subsea well operation. The intervention system includes a first coupling portion, a second coupling portion releasably coupled to the first coupling portion, a monitor tether coupled to one of the first coupling portion and the second coupling portion, and a control system. The first coupling portion and the second coupling portion provide fluid communication while the first coupling portion and the second coupling portion are connected to one another. The monitor tether transmits an electrical current while the first coupling portion and the second coupling portion are connected to one another. The control system is operable to detect a change in current in the monitor tether indicative of disconnection between the first coupling portion and the second coupling portion, and, in response to detecting a change in current in the monitor tether indicative of disconnection, interrupt fluid flow relative to the subsea well.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to offshore subsea operations, and more particularly to a disconnection system for subsea wells or tie in points.

BACKGROUND

(2) Operation of subsea drilling and intervention systems may include a disconnect system to selectively detach subsea equipment from equipment (e.g., a tree) located proximate a subsea well. Flow to/from a well may be interrupted when a disconnect event occurs. The disconnection may protect the environment and/or the drilling or intervention equipment.

SUMMARY

(3) In one independent aspect, an intervention system is provided for a subsea well operation. The intervention system includes a first coupling portion, a second coupling portion releasably coupled to the first coupling portion, a monitor tether coupled to one of the first coupling portion and the second coupling portion, and a control system. The first coupling portion and the second coupling portion are configured to provide communication between a vessel and a subsea well while the first coupling portion and the second coupling portion are connected to one another. The monitor tether is configured to transmit an electrical current while the first coupling portion and the second coupling portion are connected to one another. The control system is operable to detect a change in current in the monitor tether indicative of disconnection between the first coupling portion and the second coupling portion, and, in response to detecting a change in current in the monitor tether indicative of disconnection, interrupt fluid flow relative to the subsea well.

(4) In some aspects, the control system is configured to be positioned on a subsea safety module, wherein the monitor tether is configured to be coupled between the subsea safety module and the one of the first coupling portion and the second coupling portion.

(5) In some aspects, the system further includes a switch operable between an open state and a closed state, wherein the switch is in closed state while the first coupling portion and the second coupling portion are connected to one another, and the switch is in the open state while the first coupling portion and the second coupling portion are disconnected from one another, wherein electrical current is prevented from passing through the monitor tether while the switch is in the open state.

(6) In some aspects, the switch includes a first switch portion positioned on the first coupling portion and a second switch portion positioned on the second coupling portion, wherein the switch is in the closed state while the first switch portion and the second switch portion are in contact with one another.

- (7) In some aspects, the monitor tether is removably coupled to a portion of the switch.
- (8) In some aspects, the monitor tether is removably coupled to the control system.
- (9) In some aspects, in response to detecting a change in current indicative of disconnection between the first coupling portion and the second coupling portion, the control system is operable to actuate a valve to prevent at least one of flow into the well and flow out of the well.
- (10) In another independent aspect, an intervention system is provided for a subsea well operation. The subsea well operation including a conduit providing communication between a surface vessel and a subsea tree. The intervention system includes a releasable coupling configured to selectively interrupt fluid communication in the conduit, and a sensor for detecting the condition of the releasable coupling. The releasable coupling includes a first coupling portion and a second coupling portion. The first coupling portion is configured to be coupled to a first portion of the conduit, and the second coupling portion is configured to be coupled to a second portion of the conduit. The second coupling portion is movable between a first condition in which the second coupling portion is coupled to the first coupling portion, and a second condition in which the second coupling portion is separated from the first coupling portion. The sensor transmits a signal indicative of the condition of the second coupling portion.
- (11) In some aspects, the sensor includes a switch operable between an open state and a closed state, the switch being in closed state while the first coupling portion and the second coupling portion are connected to one another, and the switch is in the open state while the first coupling portion and the second coupling portion are disconnected from one another.
- (12) In some aspects, the switch includes a first switch portion positioned on the first coupling portion and a second switch portion positioned on the second coupling portion, wherein the switch is in the closed state while the first switch portion and the second switch portion are in contact with one another.
- (13) In some aspects, the system further includes a control system operable to determine, based on the signal indicative of the condition of the second coupling portion, whether the second coupling portion is disconnected from the first coupling portion, and in response to determining that the second coupling portion is disconnected from the first coupling portion, interrupt fluid flow through the conduit.
- (14) In some aspects, the sensor transmits the signal via wireless communication.
- (15) In some aspects, the sensor transmits the signal via a monitor tether for providing electrical communication between the sensor and a subsea safety module.
- (16) In some aspects, the system further includes a mechanical tether including a first end coupled to the first coupling portion and a second end configured to be coupled to a subsea safety module.
- (17) In yet another independent aspect, an intervention system is provided for a subsea well operation. The subsea well operation includes a conduit providing communication between a surface vessel and a subsea tree. The intervention system includes a releasable coupling configured to selectively interrupt fluid communication in the conduit, a sensor for detecting the condition of the releasable coupling, a mechanical tether, and a monitor tether coupled to one of the first coupling portion and the second coupling portion. The releasable coupling includes a first coupling portion and a second coupling portion. The first coupling portion is configured to be coupled to a first portion of the conduit, and the second coupling portion is configured to be coupled to a second portion of the conduit. The second coupling portion is movable between a first condition in which the second coupling portion is coupled to the first coupling portion, and a second condition in which the second coupling portion is separated from the first coupling portion.
- (18) The mechanical tether includes a first end coupled to the first coupling portion and a second end configured to be coupled to a subsea safety module. The monitor tether is configured to transmit an electrical current while the first coupling portion and the second coupling portion are connected to one another.
- (19) In some aspects, the sensor includes a switch operable between an open state and a closed

state, the switch being in closed state while the first coupling portion and the second coupling portion are connected to one another, and the switch is in the open state while the first coupling portion and the second coupling portion are disconnected from one another.

(20) In some aspects, the switch includes a first switch portion positioned on the first coupling portion and a second switch portion positioned on the second coupling portion, wherein the switch is in the closed state while the first switch portion and the second switch portion are in contact with one another.

(21) In some aspects, the first coupling portion is positioned on a portion of the conduit coupled to the subsea tree, and the second coupling portion is positioned on a portion of the conduit coupled to the vessel, wherein the monitor tether includes one end coupled to the first coupling portion.

(22) In some aspects, the first coupling portion is positioned on a portion of the conduit coupled to the subsea tree, and the second coupling portion is positioned on a portion of the conduit coupled to the vessel, wherein the monitor tether includes one end coupled to the second coupling portion, wherein another end of the monitor tether is configured to be removably coupled to a subsea safety module.

(23) In some aspects, the monitor tether is longer than the mechanical tether.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject disclosure is further described in the following detailed description, and the accompanying drawings and schematics of non-limiting embodiments of the subject disclosure. The features depicted in the figures are not necessarily shown to scale. Certain features of the embodiments may be shown exaggerated in scale or in somewhat schematic form, and some details of elements may not be shown in the interest of clarity and conciseness. These and other features, aspects, and advantages of the present invention will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

(2) FIG. 1 illustrates a schematic view of a subsea intervention operation;

(3) FIG. 2A illustrates a schematic view of a well intervention system according to one embodiment, with a disconnect system in a connected state;

(4) FIG. 2B illustrates a schematic view of the well intervention system of FIG. 2A with the disconnect system in a disconnected state;

(5) FIG. 3A illustrates a schematic view of a well intervention system according to another embodiment, with a disconnect system in a connected state;

(6) FIG. 3B illustrates a schematic view of the well intervention system of FIG. 3A with the disconnect system in a disconnected state;

(7) FIG. 4 illustrates a schematic view of a well intervention system according to another embodiment, with a disconnect system in a connected state;

(8) FIG. 5A illustrates a schematic view of a well intervention system according to another embodiment, with a disconnect system in a connected state; and

(9) FIG. 5B illustrates a schematic view of the well intervention system of FIG. 5A with the disconnect system in a disconnected state.

DETAILED DESCRIPTION

(10) Before any embodiments are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of the configuration and arrangement of components set forth in the following description or illustrated in the accompanying drawings. The disclosure is capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be

regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

(11) FIG. 1 depicts a subsea intervention operation, including a vessel **10** (e.g., a surface vessel) positioned substantially above a wellbore **12** positioned on a seafloor **14**. A well structure (e.g., tree **16**) is positioned above the seafloor **14** (e.g., at the mouth of the wellbore **12**) and includes an assembly of valves, spools, and fittings designed to control the flow of fluids into and/or out of the well. A safety module **18** may be integrated with or operably connected to the tree **16**. The safety module **18** is configured to perform important safety functions, such as blowout prevention and initiating emergency shutdown (e.g., in response to unsafe conditions).

(12) One or more conduits **20** may connect or provide fluid communication between the vessel **10** or equipment aboard the vessel **10** and the wellbore **12** or equipment proximate to the wellbore **12**. The conduits **20** may be provided in the form of coiled tubing or may be provided in other forms. A manifold **22** may separate, consolidate, and/or redistribute flows from the wellbore **12** and/or other wellbores **12** in the vicinity (not shown) may be in fluid communication with one or more conduits **20**. In the illustrated embodiment of FIG. 1, a primary conduit **20a** may extend between the tree **16** (e.g., the primary conduit **20a** may have an end connected to the safety module **18**) and the manifold **22**, and two or more secondary conduits **20b** may extend between the manifold **22** and the vessel **10** or equipment aboard the vessel **10**. In other embodiments, other configurations may include a different number of wellbores **12**, conduits **20**, and/or manifolds **22**.

(13) A disconnect system **24** is operable to rapidly detach subsea equipment (e.g., the manifold **22**, conduits **20a**, **20b**, etc.) from the safety module **18** and/or other equipment proximate the wellbore **12**. In the illustrated embodiment, the disconnect system **24** is positioned at an intermediate location along the primary conduit **20a**. For example, the disconnect system **24** may be a mid-line weak link (MLWL) providing a predetermined point of failure/disconnection along the load path between the vessel **10** and the wellbore **12**. The disconnect system **24** may facilitate disconnection in response to a certain condition (e.g., tension in the conduit **20a** exceeding a predetermined threshold) to avoid damage to various components of the operation. In this way, the disconnect system **24** protects the integrity of the rest of the system.

(14) As shown in FIG. 2A, in the illustrated embodiment, the disconnect system **24** may include a first or lower coupling portion **26** connected to the tree **16** (for example, via the safety module **18**) and a second or upper coupling portion **30** releasably coupled to the lower coupling portion **26**. The lower coupling portion **26** may be coupled to the safety module by a mechanical tether **28**. In some embodiments, the mechanical tether **28** is a flexible connector (e.g., a cable) that may provide high tensile strength and is resistant to fatigue and corrosion. In some embodiments, the mechanical tether **28** may be formed from metallic wire, synthetic fiber (e.g., polyester, nylon, etc.), Kevlar and aramid fibers, and/or other materials.

(15) As shown in FIGS. 2A and 2B, the disconnect system may include a switch **38** that is connected to and in communication with a disconnect monitor **40** positioned on the safety module **18**. In the illustrated embodiment, the switch **38** includes a fixed portion **42** connected to the lower coupling portion **26** and a movable portion **44** connected to the upper coupling portion **30**. The switch **38** may be a position switch, a shunt cap, or another type of device. In some embodiments, the switch **38** may be a sensor that detects whether the lower coupling portion **26** and the upper coupling portion **30** are connected to one another.

(16) During operation, the upper coupling portion **30** may become disconnected from the lower coupling portion **26** in response to a predetermined condition (e.g., a tension in the conduit **20a** exceeding a predetermined threshold, due to drift of the vessel **10**, for example). Disconnection between the upper coupling portion **30** and the lower coupling portion **26** interrupts fluid

communication between the wellbore **12** and the vessel **10** (FIG. 1). In addition, the movable portion **44** of the switch **38** may be separated from the fixed portion **42**. The switch **38** is therefore operable between a first or closed state in which the movable portion **44** and the fixed portion **42** are positioned adjacent one another (i.e., the lower coupling portion **26** and the upper coupling portion **30** remain connected to one another as shown in FIG. 2A), and a second or open state in which the movable portion **44** and the fixed portion **42** are separated (FIG. 2B).

(17) In the embodiment of FIGS. 2A and 2B, the fixed portion **42** of the switch **38** is connected to the disconnect monitor **40** via a monitor tether **36**. The monitor tether **36** provides electrical communication between the switch **38** and the disconnect monitor **40**, permitting the disconnect monitor **40** to detect whether the switch **38** is in the closed state or the open state (and thus whether the coupling portions **26**, **30** are disconnected). For example, the monitor tether **36** may be provided in the form of two or more insulated electrical cables bound together and configured to facilitate the flow of signals and/or current. The disconnect monitor **40** may communicate the state of the switch **38** to another component (e.g., the safety module **18**, a remote vehicle **32** (FIG. 1)). The monitor tether **36** may be longer than the mechanical tether **28**, and the monitor tether **36** may remain in a slackened condition even while the mechanical tether **28** is taut. The monitor tether **36** may not be subject to large loads. Nonetheless, in some embodiments, the monitor tether **36** may be mechanically reinforced or strengthened.

(18) The safety module **18** may take corrective action in response to a signal from the disconnect monitor **40** indicating that a disconnect event has occurred. In some embodiments, a control system **46** within the safety module **18** may receive signals from the disconnect monitor **40** (or the disconnect monitor **40** may permit the transmission of signals and current between the switch **38** and the control system **46**). The control system **46** may be configured to take corrective action in response to one or more signals from the switch **38** by operating one or more actuators **48** to close one or more valves **50** positioned along a flowline **52** that extends through the safety module **18** and provides fluid communication between the tree **16** and the primary conduit **20a**. In this way, the safety module **18** may seal the tree **16** and interrupt flow into or out of the wellbore **12** (FIG. 1). One or more lines **54** may supply power to and/or may facilitate electrical communication between the disconnect monitor **40**, the control system **46**, the actuators **48**, and the valves **50**. Alternatively, the safety module **18** may take corrective action in another manner.

(19) In the embodiment depicted in FIGS. 2A and 2B, the disconnect monitor **40** is positioned between the switch **38** and the control system **46** and is configured to provide continuity between the switch **38** and the control system **46**. This facilitates convenient replacement or maintenance of the monitor tether **36** in the event that the monitor tether **36** is damaged. However, in other embodiments, the disconnect monitor **40** may be omitted and the monitor tether **36** may be a fixed cable (e.g., fixedly coupled to the safety module **18**) directly connected to and in communication with the line **54** connected to the control system **46**.

(20) As shown in FIG. 2A, while the switch **38** is in the closed state, the fixed portion **42** and movable portion **44** may be in close proximity (e.g., in contact with one another) such that a closed circuit is formed and an electric current passes continuously through the switch **38**. The disconnect monitor **40** may be configured to detect the presence or absence of the electric current within the switch **38**. In the illustrated embodiment, the control system **46** determines that the coupling portions **26**, **30** remain coupled to one another as long as electric current is detected passing through the switch **38**.

(21) On the other hand, as shown in FIG. 2B, when a disconnect event occurs, the upper coupling portion **30** may pull the movable portion **44** of the switch **38** away from the fixed portion **42** (e.g., in the direction **56**), thereby moving the switch **38** to the open state and opening the circuit. Thus, the flow of electric current through the switch **38** may cease and the disconnect monitor **40** detects the difference in the current (e.g., a reduced current or absence of current). Accordingly, the control system **46** determines that the coupling portions **26**, **30** are disconnected from one another.

Providing consistent electrical connection or communication between the coupling portions **26, 30** and the safety module **18** via the monitor tether **36** facilitates constant monitoring of the disconnect system **24** and more responsive (e.g., automatic) shutdown of flow through the tree **16**, safety module **18**, and/or conduits **20** when a disconnect event occurs.

(22) The disconnect system may be integrated in an operation in which fluid is directly injected into the well through the tree. In other embodiments, the disconnect system may be integrated in an operation in which fluid is injected into the well via a jumper coupling.

(23) FIGS. **3A** and **3B** illustrate a disconnect system **224** according to another embodiment. The disconnect system **224** includes several of the same components as the disconnect system **24**. Components of the disconnect system **224** and components of the disconnect system **24** having similar names and reference numbers with a difference of 200 may be substantially similar in form and function. The disconnect system **224** of FIGS. **3A** and **3B** is substantially similar to the disconnect system **24**, except that a switch **238** is positioned on the safety module **18**. A fixed portion **242** of the switch **238** may be positioned on or inside the safety module **18**, and a movable portion **244** may be releasably connected thereto. The movable portion **244** may be connected to the fixed portion **242** while the switch **238** is in the closed state (FIG. **3A**) and the movable portion **244** may be disconnected from the fixed portion **242** when the switch **238** is in the open state (FIG. **3B**). A monitor tether **236** may be connected to the movable portion **244** at one end and to the upper coupling portion **30** at another end (e.g., via an attachment member **258**).

(24) While the switch **238** is in the closed configuration, a closed circuit is provided and a current passes through the monitor tether **236** and between the fixed portion **242** and the movable portion **244**. The presence or absence of current passing between the fixed portion **242** and the movable portion **244** is detected and communicated to the control system **46** (e.g., via lines **54**). If, for example, the primary conduit **20a** is placed under an excessive tension or exceeds another threshold condition, a disconnect event occurs and the upper coupling portion **30** separates from the lower coupling portion **26**. As the upper coupling portion **30** moves (e.g., in the direction **56** shown in FIG. **3B**), the monitor tether **236** becomes taut and pulls the movable portion **244** of the switch **238** out of contact with the fixed portion **242**, thereby transitioning the switch **238** to the open state and terminating the current running between the fixed portion **242** and the movable portion **244**. In response to detecting a change in current (e.g., a reduction and/or absence of current), the control system **46** determines that a disconnect event has occurred and shuts off the flowline **52** (e.g., by closing the valves **50** via the actuators **48**).

(25) FIG. **4** illustrates a disconnect system **424** according to another embodiment. The disconnect system **424** includes several of the same components as the disconnect system **24**. Components of the disconnect system **424** and components of the disconnect system **24** having similar names and reference numbers with a difference of 400 may be substantially similar in form and function. The disconnect system **424** of FIG. **4** is substantially similar to the disconnect system **24**, except that communication between a sensor **438** and a safety module **18** is wireless. For example, the sensor **438** may be configured to send electronic signals to the safety module **18** using wireless communication protocols including Wi-Fi, Bluetooth, subsea acoustics, near field communication (NFC), Zigbee, cellular networks, LoRa, and the like. The sensor **438** may be configured to send wireless signals to the safety module **18**. In the illustrated embodiment, the sensor **438** communicates with a disconnect monitor **440**. In other embodiments, the sensor **438** may communicate wirelessly with the safety module **18** directly (e.g., by sending signals directly to the control system **46**).

(26) The sensor **438** may include a switch. While the coupling portions **26, 30** are connected to one another, the switch is in the closed configuration and a closed circuit is provided between a fixed portion **442** of the switch and a movable portion **444** of the switch. When a disconnect event occurs, the switch transitions to an open state and the circuit is broken, triggering a change in the wireless signal transmitted from the sensor **438**. The control system **46** (e.g., via the disconnect

monitor **440**) detects the change in signal. When the control system **46** detects a change that is indicative of a disconnection, the control system **46** determines that a disconnect event has occurred and shuts off the flowline **52** (e.g., by closing the valves **50** via the actuators **48**).

(27) FIGS. **5A** and **5B** illustrate a disconnect system **524** according to another embodiment. The disconnect system **524** is substantially similar to the disconnect system **24** depicted in FIGS. **2A** and **2B**, and like parts with reference numbers having a difference of 500 may be substantially similar in form and function. A control line **558** extends between the disconnect system **524** and the surface or the vessel **10** (FIG. **1**). The control line **558** is routed from the surface to the control system **46** via one or both of a disconnect system **524** and a switch **538** so that a disconnect severs power and communication to the control system **46**, thereby triggering operation of the actuators **48** to close the valves **50**. For example, the control line **558** may be connected in line with the monitor tether **536** such that electrical communication is provided between the control line **558** and the monitor tether **536**. In some embodiments, the control line **558** may be connected to the switch **538** (e.g., to the movable portion **544** of the switch **538**) such that the control line **558** provides communication between the switch **538** and the surface and the monitor tether **536** provides communication between the control line **558** and the safety module **18**. Thus, if a disconnect occurs when the disconnect system **524** is in communication with the surface via the control line **558**, the control system **46** may operate the actuators **48** to close the valves **50**.

(28) In other embodiments, other configurations are possible. For example, those of skill in the art will recognize, according to the principles and concepts disclosed herein, that various combinations, sub-combinations, and substitutions of the components discussed above can provide a mud pulse telemetry system.

(29) The embodiment(s) described above and illustrated in the figures are presented by way of example only and are not intended as a limitation upon the concepts and principles of the present disclosure. As such, it will be appreciated that variations and modifications to the elements and their configuration and/or arrangement exist within the spirit and scope of one or more independent aspects as described.

Claims

1. An intervention system for a subsea well operation, the intervention system comprising: a first coupling portion; a second coupling portion releasably coupled to the first coupling portion, the first coupling portion and the second coupling portion configured to provide communication between a vessel and a subsea well while the first coupling portion and the second coupling portion are connected to one another; a monitor tether coupled to one of the first coupling portion and the second coupling portion, the monitor tether configured to transmit an electrical current while the first coupling portion and the second coupling portion are connected to one another; and a control system operable to detect a change in current in the monitor tether indicative of disconnection between the first coupling portion and the second coupling portion, and in response to detecting the change in the current in the monitor tether indicative of the disconnection, interrupt fluid flow relative to the subsea well.
2. The system of claim 1, wherein the control system is configured to be positioned on a subsea safety module, wherein the monitor tether is configured to be coupled between the subsea safety module and the one of the first coupling portion and the second coupling portion.
3. The system of claim 1, further comprising a switch operable between an open state and a closed state, wherein the switch is in the closed state while the first coupling portion and the second coupling portion are connected to one another, and the switch is in the open state while the first coupling portion and the second coupling portion are disconnected from one another, wherein the electrical current is prevented from passing through the monitor tether while the switch is in the open state.

4. The system of claim 3, wherein the switch includes a first switch portion positioned on the first coupling portion and a second switch portion positioned on the second coupling portion, wherein the switch is in the closed state while the first switch portion and the second switch portion are in contact with one another.
5. The system of claim 4, wherein the monitor tether is removably coupled to a portion of the switch.
6. The system of claim 3, wherein the monitor tether is removably coupled to the control system.
7. The system of claim 1, wherein, in response to detecting the change in the current indicative of the disconnection between the first coupling portion and the second coupling portion, the control system is operable to actuate a valve to prevent at least one of flow into the well and flow out of the well.
8. An intervention system for a subsea well operation, the subsea well operation including a conduit providing communication between a surface vessel and a subsea tree, the intervention system comprising: a releasable coupling configured to selectively interrupt fluid communication in the conduit, the releasable coupling including a first coupling portion and a second coupling portion, the first coupling portion configured to be coupled to a first portion of the conduit, the second coupling portion configured to be coupled to a second portion of the conduit, the second coupling portion movable between a first condition in which the second coupling portion is coupled to the first coupling portion, and a second condition in which the second coupling portion is separated from the first coupling portion; and a sensor for detecting the condition of the releasable coupling, the sensor transmitting a signal indicative of the condition of the second coupling portion, wherein the sensor transmits the signal via wireless communication or transmits the signal via a monitor tether for providing electrical communication between the sensor and a subsea safety module.
9. The system of claim 8, wherein the sensor includes a switch operable between an open state and a closed state, the switch being in the closed state while the first coupling portion and the second coupling portion are connected to one another, and the switch is in the open state while the first coupling portion and the second coupling portion are disconnected from one another.
10. The system of claim 9, wherein the switch includes a first switch portion positioned on the first coupling portion and a second switch portion positioned on the second coupling portion, wherein the switch is in the closed state while the first switch portion and the second switch portion are in contact with one another.
11. The system of claim 8, further comprising a control system operable to determine, based on the signal indicative of the condition of the second coupling portion, whether the second coupling portion is disconnected from the first coupling portion, and in response to determining that the second coupling portion is disconnected from the first coupling portion, interrupt fluid flow through the conduit.
12. The system of claim 8, further comprising a mechanical tether including a first end coupled to the first coupling portion and a second end configured to be coupled to the subsea safety module.
13. An intervention system for a subsea well operation, the subsea well operation including a conduit providing communication between a surface vessel and a subsea tree, the intervention system comprising: a releasable coupling configured to selectively interrupt fluid communication in the conduit, the releasable coupling including a first coupling portion and a second coupling portion, the first coupling portion configured to be coupled to a first portion of the conduit, the second coupling portion configured to be coupled to a second portion of the conduit, the second coupling portion movable between a first condition in which the second coupling portion is coupled to the first coupling portion, and a second condition in which the second coupling portion is separated from the first coupling portion; a sensor for detecting the condition of the releasable coupling; a mechanical tether including a first end coupled to the first coupling portion and a second end configured to be coupled to a subsea safety module; and a monitor tether coupled to one of the first coupling portion and the second coupling portion, the monitor tether configured to

transmit an electrical current while the first coupling portion and the second coupling portion are connected to one another.

14. The system of claim 13, wherein the sensor includes a switch operable between an open state and a closed state, the switch being in the closed state while the first coupling portion and the second coupling portion are connected to one another, and the switch is in the open state while the first coupling portion and the second coupling portion are disconnected from one another.

15. The system of claim 14, wherein the switch includes a first switch portion positioned on the first coupling portion and a second switch portion positioned on the second coupling portion, wherein the switch is in the closed state while the first switch portion and the second switch portion are in contact with one another.

16. The system of claim 13, wherein the first coupling portion is positioned on a portion of the conduit coupled to the subsea tree, and the second coupling portion is positioned on a portion of the conduit coupled to the vessel, wherein the monitor tether includes one end coupled to the first coupling portion.

17. The system of claim 13, wherein the first coupling portion is positioned on a portion of the conduit coupled to the subsea tree, and the second coupling portion is positioned on a portion of the conduit coupled to the vessel, wherein the monitor tether includes one end coupled to the second coupling portion, wherein another end of the monitor tether is configured to be removably coupled to the subsea safety module.

18. The system of claim 13, wherein the monitor tether is longer than the mechanical tether.

19. The system of claim 13, wherein a control line is routed from the surface vessel to the subsea tree via the releasable coupling such that, when the second coupling portion moves from the first condition to the second condition, a control system within the subsea tree operates one or more actuators to close off one or more valves.
